Genome-wide association studies test genetic variants against traits

Quantitative and binary models, diagnostics, and evidence thresholds

Kam

Tulane University | Biomedical Informatics

A regression for each variant — then evaluate the evidence across the genome.

Quantitative-trait GWAS models a change in a measured outcome

Example: body mass index (BMI), a continuous phenotype

$$\mathrm{BMI}_i = \beta_0 + \beta_G G_{ij} + \gamma_1 \mathrm{Age}_i + \gamma_2 \mathrm{Sex}_i + \varepsilon_i$$

Genotype	G dosage	Illustrative BMI
AA	0	24
AG	1	27
GG	2	30

Why code 0, 1, or 2?

- A diploid person carries two allele copies at an autosomal SNP.
- The additive model estimates the average change *per additional G allele*.
- Other inheritance models can be used when justified.

If $\widehat{\beta}_G = 1.5$, the fitted mean BMI differs by 1.5 units per G copy after adjustment. The table is a teaching example, not a study result.

Adapted from BIMI 6400 Module 10, pp. 7–8; dosage example is illustrative.

G_{ij} is the coded allele dosage for person i at SNP j

Read the subscripts before fitting the regression

i identifies the participant; j identifies the variant. G_{ij} records how many copies of the designated effect allele that person carries at that SNP.

Person i	SNP j	Genotype	G_{ij}
01	rsX	AA	0
02	rsX	AG	1
03	rsX	GG	2

When a genotype is imputed

$$G_{ij} = 0P(AA) + 1P(AG) + 2P(GG).$$

For probabilities 0.10, 0.20, 0.70:

$$G_{ij} = 0 + 0.20 + 1.40 = 1.60.$$

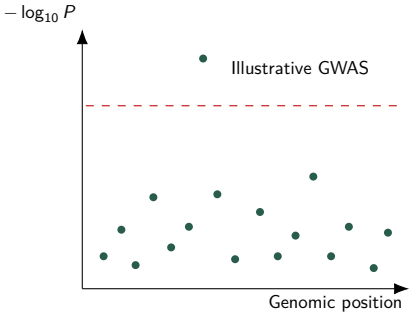
The effect allele must be specified. If we change the coded allele from G to A, the dosage and regression-effect direction reverse.

BIMI 6400 Module 10, p. 8; genotype-imputation expansion is added teaching material.

A genome-wide scan repeats the regression at each tested SNP

Same phenotype, successive SNPs

Test	Variant	P-value
1	rsA	0.40
2	rsB	2×10^{-9}
3	rsC	0.002
⋮	⋮	⋮



Each row is a *different per-variant regression*, adjusted for the planned covariates.

Insert your own figure: save it as `gwas_figure.png` beside this `.tex` file; the illustration will be replaced automatically.

BIMI 6400 Module 10, pp. 8–9, 16. Plot and table illustrative.

Binary-trait GWAS models the odds of disease, not the trait mean

Example: asthma (case = 1, control = 0)

$$\log\left(\frac{\Pr(Y_i = 1)}{1 - \Pr(Y_i = 1)}\right) = \beta_0 + \beta_G G_{ij} + \boldsymbol{\gamma}^\top \mathbf{X}_i.$$

Why logistic regression?

- A case/control outcome is coded 0 or 1.
- A linear mean model may predict probabilities outside $[0, 1]$.
- Logistic regression relates genotype to *log odds*.

Interpret an effect If $\widehat{\beta}_G = \log(1.5)$, then $e^{\widehat{\beta}_G} = 1.5$. Each extra effect allele is associated with **50% higher odds** of asthma after adjustment—not 50% higher probability.

A participant's observed diagnosis and their genomic genotype are different types of data; both must be defined correctly.

BIMI 6400 Module 10, pp. 7–8; quantitative/binary distinction expanded.

LD patterns differ because populations have different genetic histories

Linkage disequilibrium (LD)

LD is the non-random co-occurrence of alleles at two genomic positions. The same marker may tag a nearby causal allele well in one population and less well in another.

Two illustrative haplotype patterns

Haplotype	Pop. A	Pop. B
AB	.45	.30
Ab	.05	.20
aB	.05	.20
ab	.45	.30

Why do these differences arise?

- Historical recombination breaks up ancestral haplotypes.
- Genetic drift, bottlenecks, founder events and admixture alter frequencies.
- Local allele frequency and sample history shape measured r^2.

Both populations have $p(A) = p(B) = 0.5$, yet r^2 differs: **0.64 in A, 0.04 in B**. This demonstrates why frequency alone does not describe LD.

Extended teaching explanation; Korte & Farlow, 2013, pp. 3–5 (LD and population structure). Hypothetical frequencies.

Array haplotypes let us infer variants that were not directly typed

An array measures selected markers; imputation uses reference haplotypes

Nearby SNPs are inherited together in recognizable haplotype patterns. A reference panel of phased chromosomes helps predict alleles at variants missing from the genotyping array.

Example haplotype	Typed 1	Missing	Typed 2
H1	A	**G**	C
H2	T	**A**	G

If an individual's typed pattern resembles H1, the untyped middle allele may be predicted as G.

Important limits

- Prediction is probabilistic, not sequencing.
- Reference-panel representation affects accuracy.
- Low-frequency variants can be difficult to impute reliably.

Connection to dosage: imputation returns genotype probabilities that can be summarized as a 0–2 expected effect-allele count.

Korte & Farlow, 2013, p. 4 (imputation); genomic-array explanation added for classroom use.

GWAS requires correctly linked files and quality-controlled measurements

Input files: one participant ID system

Input	Contains
Genotype	Variant/dosage data
Phenotype	BMI or asthma status
Keep	Eligible participant IDs
Covariates	Age, sex, PCs, site

Check before regression

Domain	Example check
Genotypes	Call rate, imputation INFO
Samples	ID, relatives, ancestry
Phenotype	Case counts, coding
Model	Covariate missingness

Common failure: the genotype file uses one sample ID format and the phenotype file another; the join silently drops participants.

BIMI 6400 Module 10, p. 21; QC details are additional teaching material.

A GWAS runs many regressions and tests one SNP coefficient in each

The test performed at each SNP j

$$H_0 : \beta_{G_j} = 0 \qquad \text{versus} \qquad H_1 : \beta_{G_j} \neq 0$$

Outcome	**Typical model**	**Effect estimate**
BMI (quantitative)	Linear regression	Change in mean BMI per allele
Asthma (binary)	Logistic regression	Odds ratio per allele

Every fitted test reports an effect estimate, a standard error, and a P-value. **One SNP with $P < 0.05$ is not enough evidence** after hundreds of thousands or millions of tests.

BIMI 6400 Module 10, pp. 7–10.

At $P < 0.05$, thousands of null SNPs produce false alarms

What does a 5% false-positive rate mean?

If a SNP has no association, a valid test rejects its true null hypothesis with probability 0.05. When we test many true-null SNPs, some pass by chance.

True-null SNP tests	Expected false positives at 0.05	Expected at 0.001
100	5	0.1
1,000	50	1
10,000	500	10

This is an *expected number*, not a guaranteed count. Correlation among SNP tests changes the distribution of errors, though the expectation follows from valid per-test error rates.

BIMI 6400 Module 10, pp. 10–11; numerical explanation added.

Family-wise error means the chance of at least one false positive

Imagine a study where all tested SNPs have no real effect

Under *independent* true-null tests, the probability of no false positives across M tests is $(1-\alpha)^M$.

$$P(\text{one or more false positives}) = 1-(1-\alpha)^M$$

Tests M	No false positives	At least one false positive
1	95%	5%
10	$\approx$ 60%	$\approx$ 40%
100	$\approx$ 0.6%	$\approx$ 99.4%

Why correction matters: a significance level chosen for one test does not protect an entire genome scan. Real SNP tests are correlated by LD, so this table is an intuition-building illustration.

BIMI 6400 Module 10, pp. 10–11.

Bonferroni is conservative; FDR answers a different question

Method	What it seeks to control	How to think about it
Bonferroni	Chance of *any* false positive (FWER)	Test $P \leq 0.05/M$
Holm	Chance of *any* false positive (FWER)	Step-down correction; at least as powerful as Bonferroni
Benjamini–Hochberg	Expected proportion of false discoveries (FDR)	Orders P-values; commonly less restrictive

Example: one million tests

Bonferroni yields $0.05/1{,}000{,}000 = 5 \times 10^{-8}$. BH at FDR $q = 0.05$ is **not** the same cutoff and can allow more discoveries.

Bonferroni is the most conservative of these three standard procedures for the same family of tests; the preferred method depends on whether discovery or strict false-positive control is the goal.

BIMI 6400 Module 10, p. 12; Holm and FDR comparison added for teaching.

A GWAS table reports effect size, precision, and evidence

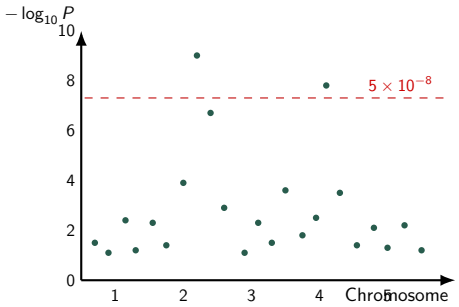
Illustrative output for a quantitative trait

SNP	$\widehat{\beta}_G$	*SE*	*P*-**value**
rsA	0.02	0.04	0.62
rsB	0.10	0.02	5.7×10^{-7}
rsC	0.18	0.03	2.0×10^{-9}

- $\widehat{\beta}_G$: fitted difference in the trait per effect allele.
- *SE*: precision of the fitted estimate; smaller values usually imply greater precision.
- *P*: strength of evidence against $\beta_G = 0$, not the probability that a SNP causes disease.

BIMI 6400 Module 10, p. 9. Table values are illustrative.

A Manhattan plot shows where association signals cluster



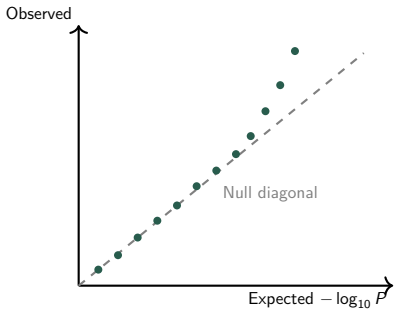
Read the plot

- Each dot represents one tested SNP.
- Higher dot means smaller P.
- A group of nearby high dots often reflects LD.
- The horizontal dashed line is a chosen significance threshold.

A peak points to a genomic region. **It does not, by itself, identify the causal variant or gene.**

BIMI 6400 Module 10, p. 16. Entire plot is schematic (5 chromosomes shown for clarity).

A Q–Q plot compares observed P-values with the null distribution



It is not a regression-residual plot

- Horizontal: expected P-value quantiles under the null.
- Vertical: observed SNP P-value quantiles.
- Diagonal: good agreement with null expectations.
- Broad upward deviation: inspect confounding, relatedness, batch effects, and polygenicity.

Defensible interpretation: read Q–Q and Manhattan plots alongside phenotype validity, ancestry adjustment, case counts, and genotype QC. A clean plot is not proof of causality.

BIMI 6400 Module 10, pp. 17–19; additional diagnostic explanation.

The standard 5×10^{-8} threshold comes from common-variant GWAS

Fadista et al. (2016): reason to revisit it

The usual cutoff grew from common-variant studies. Sequencing reveals many additional low-frequency variants, and the number of effectively distinct tests depends on which variants are included.

Analysis	**Main consideration**
Common SNP array	Many linked markers; conventional genome-wide threshold often used
Whole-exome sequence	Variants in protein-coding regions; smaller genomic target
Whole-genome sequence	Coding and noncoding variants; much broader testing burden

The number of *independent or nearly independent* SNP tests, not only the raw number genotyped, affects the cutoff.

Fadista et al., Eur J Hum Genet 2016, Abstract and Introduction.

The number of distinct variants determines a study's testing burden

Why LD changes a multiple-testing correction

Markers inherited together can yield highly correlated tests. LD pruning retains representative markers and provides an approximate count of distinct tests.

$$\alpha_{\text{per test}} \approx \frac{0.05}{M_{\text{eff}}}$$

Illustrative number of effective tests	Approximate cutoff	Interpretation
500,000	1×10^{-7}	Less stringent
1,000,000	5×10^{-8}	Conventional benchmark
5,000,000	1×10^{-8}	More stringent

More nearly independent tests require smaller P-values for the same target family-wise error rate.

Fadista et al., 2016, Methods. Table uses illustrative counts, not measured results.

GoT2D and T2D-GENES are study projects, not single disease genes

The paper used genetic data collected by diabetes research consortia

Name	Plain-language meaning	How used in the paper
GoT2D	Genetics of Type 2 Diabetes	Genomes/exomes from 2,875 European-ancestry participants
T2D-GENES	Type 2 Diabetes sequencing collaboration	Combined exome study: about 13,000 people from five ancestry groups

The authors were **estimating the multiple-testing burden**, not reporting a new asthma or diabetes risk allele. Sequencing data are being used to count distinguishable variant tests.

Fadista et al., 2016, Abstract and Methods.

Including lower-frequency variants leads to more stringent proposed cutoffs

Proposed European-ancestry significance levels from Fadista et al.

Included variant frequency	Whole genome	Whole exome
MAF $\geq 5\%$	5×10^{-8}	1×10^{-6}
MAF $\geq 0.5\%$	2×10^{-8}	5×10^{-7}
MAF $\geq 0.1\%$	1×10^{-8}	3×10^{-7}

Example: a whole-genome result with $P = 3 \times 10^{-8}$ passes 5×10^{-8} but **not** the proposed 2×10^{-8} threshold for MAF $\geq 0.5\%$. These are this article's proposed study-specific thresholds, not universal rules for every modern dataset.

Fadista et al., 2016, Results/Discussion; LD criterion $r^2 < 0.8$.

Different LD patterns can change which P-values clear a threshold

Fadista et al.: an ancestry-related illustration

For whole-genome variants with MAF $\geq 0.5\%$ (LD criterion $r^2 < 0.8$):

Study subset	**Paper's proposed cutoff**	**Does $P = 2.4 \times 10^{-8}$ pass?**
Finnish	2.6×10^{-8}	Yes
Swedish/British	2.3×10^{-8}	No

Interpretation: different demographic histories can alter LD and the number of distinct variants. The threshold should be selected in advance from the study design; it should not be changed to rescue a desired SNP. These values describe the populations and analytic filters studied in this paper, not a fixed rule for any individual's ancestry.

Fadista et al., 2016, Results and Discussion. DOI: 10.1038/ejhg.2015.269.